

Chapter 2

HTML

Basic example of HTML:

<!DOCTYPE html> The <!DOCTYPE html> declaration defines this document to be HTML5

<html> The <html> element is the root element of an HTML page

<head> The <head> element contains meta information about the document

<title>Page Title</title> The <title> element specifies a title for the document

</head>

<body> The <body> element contains the visible page content

<h1>My First Heading</h1> The <h1> element defines a large heading

<p>My first paragraph.</p> The <p> element defines a paragraph

</body>

</html>

HTML tags

- Element names surrounded by angle brackets:

`<tagname>content goes here...</tagname>`

- HTML tags normally come in pairs like `<p>` and `</p>`
- The first tag in a pair is the start tag, the second tag is the end tag
- The end tag is written like the start tag, but with a forward slash inserted before the tag name

HTML Headings

- HTML headings are defined with the `<h1>` to `<h6>` tags.
- `<h1>` defines the most important heading.
- `<h6>` defines the least important heading:

HTML Block and Inline Elements

Block-level Elements

- A block-level element always starts on a new line and takes up the full width available (stretches out to the left and right as far as it can).
- Block elements appear on the screen as if they have a line break before and after them.

For example

`<p>`, `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, `<h6>`, `<ul>`, `<ol>`, `<dl>`, `<pre>`, `<hr />`, `<blockquote>`, and `<address>` elements are all block level elements.

- They all start on their own new line, and anything that follows them appears on its own new line.

Inline Elements:

- An inline element does not start on a new line and it only takes up as much width as necessary.
- It can appear within sentences and do not have to appear on a new line of their own.
- The `<b>`, `<i>`, `<u>`, `<em>`, `<strong>`, `<sup>`, `<sub>`, `<big>`, `<small>`, `<li>`, `<ins>`, `<del>`, `<code>`, `<cite>`, elements are all inline elements

HTML TEXT FORMATTING

Tag	Purpose
 and 	To Bold Text
<i> and 	To italic Text
<sub>	Defines subscripted text
<sup>	Defines superscripted text
<small>	Defines smaller text
<big>	Defines bigger text
<ins>	Defines inserted text
	Defines deleted text
<mark>	Defines marked/highlighted text

HTML TEXT FORMATTING

1. The **** and **** tags

The **** tag is used to write bold text in HTML. The **** tag does the same, by adding a bit more emphasis. Simply put, it adds semantic importance to the text.

2. The **<i>** and **** tags

The **<i>** tag is used to make your text italic. The **** tag does the same, but by giving your italic text, semantic importance.

Basically, both the **** and **** tags appear the same as **<i>** and **** on your browser display. However, they don't mean the same things. One is used for simply making the text italic or bold, the other is used for adding emphasis on the text.

3. The **<small>** and **<big>** tags

The **<small>** tag is used to make the text appear small. The **<big>** tag does the opposite.

4. The **<mark>** tag

Another element, which is rarely used but useful all the same. The **<mark>** tags marks your text with a yellow marker to highlight.

5. The tag

A useful tag for showing removed or deleted text. It displays text with a line crossed through the middle of it.

6. The <ins> tag

Used for displaying inserted or added text. It underlines the text to highlight the fact that its been added in the middle of a line or a paragraph.

7. The <sub> and <sup> tags

The <sub> tag is used for displaying subscripts, the kind that you would probably see in chemical equations. The <sup> script on the other hand, displays the superscripts. Its most commonly used in displaying mathematical equations with power symbols.

8.Strike tag

The HTML <strike> tag gives text the appearance of a strikethrough which draws a horizontal line over the text. This tag is also commonly referred to as the <strike> element.

Note: The <strike> tag has been removed in HTML5. You can try using the [tag](#) instead.

HTML comments

- This element is used to add a comment to an HTML document.
- An HTML comment begins with `<!--` and the comment closes with `-->`.
- HTML comments are visible to anyone that views the page source code, but are not rendered when the HTML document is rendered by a browser.
- **Note:** Comments are not displayed by the browser, but they can help document your HTML source code.

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<!-- This is a comment -->
```

```
<p>This is a paragraph.</p>
```

```
<!-- Comments are not displayed in the browser -->
```

```
</body>
```

```
</html>
```

This is a paragraph

Horizontal row tag

<hr> tag

```
<!DOCTYPE html>
<html>
<body>
<p>The hr tag defines a horizontal rule:</p>
<hr>
<p>This is a paragraph.</p>
<hr>
<p>This is a paragraph.</p>
<hr>
<p>This is a paragraph.</p>
</body>
</html>
```

The hr tag defines a horizontal rule:

This is a paragraph.

This is a paragraph.

This is a paragraph.

Meta tag:

- additional important information about a document in a variety of ways. The META elements can be used to include name/value pairs describing properties of the HTML document, such as author, expiry date, a list of keywords, document author etc.
- The <meta> tag is used to provide such additional information. This tag is an empty element and so does not have a closing tag but it carries information within its attributes.
- You can include one or more meta tags in your document based on what information you want to keep in your document but in general, meta tags do not impact physical appearance of the document so from appearance point of view, it does not matter if you include them or not.
- Adding Meta Tags to Your Documents

You can add metadata to your web pages by placing <meta> tags inside the header of the document which is represented by <head> and </head> tags.

HTML LISTS

HTML offers web authors three ways for specifying lists of information. All lists must contain one or more list elements. Lists may contain –

- ** – An unordered list. This will list items using plain bullets.
- ** – An ordered list. This will use different schemes of numbers to list your items.
- <dl>** – A definition list. This arranges your items in the same way as they are arranged in a dictionary.

HTML Unordered Lists

An unordered list is a collection of related items that have no special order or sequence. This list is created by using HTML tag. Each item in the list is marked with a bullet.

```
<html>
  <head>
    <title>HTML Unordered List</title>
  </head>
  <body>
    <ul>
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ul>
  </body>
</html>
```

Output:

- Beetroot
- Ginger
- Potato
- Radish

The type Attribute:

- You can use **type** attribute for tag to specify the type of bullet you like. By default, it is a disc. Following are the possible options –

<ul type = "**square**">

<ul type = "**disc**">

<ul type = "**circle**">

The type Attribute:

- `<ul type = "square">`

`html>`

`<head>`

`<title>HTML Unordered List</title>`

`</head>`

`<body>`

`<ul type = "square">`

`<li>Beetroot</li>`

`<li>Ginger</li>`

`<li>Potato</li>`

`<li>Radish</li>`

`</ul>`

`</body>`

`</html>`

OUTPUT:

- Beetroot
- Ginger
- Potato
- Radish

The type Attribute:

<ul type = "disc">

<html>

<head>

<title>HTML Unordered List</title>

</head>

<body>

<ul type = "disc">

Beetroot

Ginger

Potato

Radish

</body>

OUTPUT

- Beetroot
- Ginger
- Potato
- Radish

The type Attribute:

```
<ul type = "circle">
```

```
<html>
```

```
<head>
```

```
<title>HTML Unordered List</title>
```

```
</head>
```

```
<body>
```

```
<ul type = "circle">
```

```
<li>Beetroot</li>
```

```
<li>Ginger</li>
```

```
<li>Potato</li>
```

```
<li>Radish</li>
```

```
</ul>
```

```
</body>
```

```
</html>
```

OUTPUT

- Beetroot
- Ginger
- Potato
- Radish

HTML Ordered Lists

```
<html>

  <head>
    <title>HTML Ordered
List</title>
  </head>

  <body>
    <ol>
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ol>
  </body>

</html>
```

OUTPUT

1. Beetroot
2. Ginger
3. Potato
4. Radish

The type Attribute:

- `<ol type = "1">` - Default-Case Numerals.
- `<ol type = "I">` - Upper-Case Numerals.
- `<ol type = "i">` - Lower-Case Numerals.
- `<ol type = "A">` - Upper-Case Letters.
- `<ol type = "a">` - Lower-Case Letters.

The start Attribute

- `<ol type = "1" start = "4">` - Numerals starts with 4.
- `<ol type = "I" start = "4">` - Numerals starts with IV.
- `<ol type = "i" start = "4">` - Numerals starts with iv.
- `<ol type = "a" start = "4">` - Letters starts with d.
- `<ol type = "A" start = "4">` - Letters starts with D.

HTML Definition Lists

- HTML supports a list style which is called definition lists where entries are listed like in a dictionary or encyclopedia.
- The definition list is the ideal way to present a glossary, list of terms, or other name/value list.

Definition List makes use of following three tags.

<dl> – (definition list)Defines the start of the list

<dt> – (define term)A term

<dd> – (definition description)Term definition

</dl> – Defines the end of the list

HTML Definition Lists

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Definition List</title>
  </head>

  <body>
    <dl>
      <dt><b>HTML</b></dt>
      <dd>This stands for Hyper Text Markup Language</dd>
      <dt><b>HTTP</b></dt>
      <dd>This stands for Hyper Text Transfer Protocol</dd>
    </dl>
  </body>
</html>
```

OUTPUT:

HTML

This stands for Hyper Text Markup Language

HTTP

This stands for Hyper Text Transfer Protocol

Nested LIST

Example 1

- Fruit
 - Bananas
 - Apples
 - Green
 - Red
 - Pears
- Vegetables
- Meat

Example 1 Solution

```
<ul type="disc">
  <li>Fruit </li>
    <ul type="circle">
      <li>Bananas</li>
      <li>Apples </li>
        <ul type="square">
          <li>Green</li>
          <li>Red</li>
        </ul>
      <li>Pears</li>
    </ul>

  <li>Vegetables</li>
  <li>Meat</li>
</ul>
```

Example 2

A list in a list in a list

- Coffee
- Tea
 - Black tea
 - Green tea
 - China
 - Africa
- Milk

Example 2 solution

```
<html>
<body>
<h1>A list in a list in a list</h1>
<ul>
  <li>Coffee</li>
  <li>Tea
    <ul type="circle">
      <li>Black tea</li>
      <li>Green tea</li>
      <ul type="square">
        <li>China</li>
        <li>Africa</li>
      </ul>
    </ul>

    </li>

    <li>Milk</li>
  </ul>

</body>
</html>
```

Example 3

1. codes
2. cracker
3. html
4. list
5. tutorial
6. examples

Example 4

1. **html**

2. **list**

A. **html**

B. **list**

a. **html**

b. **list**

I. **html**

II. **list**

i. **html**

ii. **list**

Practice Example-

List of available courses

- Data Structures & Algorithm
- Web Technology
- Aptitude & Logical Reasoning
- Programming Languages

Data Structures topics

1. Array
2. Linked List
3. Stacks
4. Queues
5. Trees
6. Graphs

Practice Example-

reversed attribute

3. HTML
2. CSS
1. JS

start attribute

5. HTML
6. CSS
7. JS

type attribute

- i. HTML
- ii. CSS
- iii. JS

HTML Description List

A Description List

Coffee

- 500 gms

Milk

- 1 ltr Tetra Pack

HTML Links - Hyperlinks

- HTML links are hyperlinks.
- we can click on a link and jump to another document.
- When we move the mouse over a link, the mouse arrow will turn into a little hand.
- HTML Links - Syntax
- Hyperlinks are defined with the HTML `<a>` tag:
- `<a href="url">link text</a>`

```
<!DOCTYPE html>
<html>
<head>
...
</head>
<body>
  ...
  <a href="url">text</a>
  ...
</body>
</html>
```



```
<!DOCTYPE html>
<html>
<body>

<h2>HTML Links</h2>
<p><a href="https://www.w3schools.com/html/">Visit our HTML tutorial</a></p>

</body>
</html>
```

HTML Links

[Visit our HTML tutorial](https://www.w3schools.com/html/)

- The **href** attribute specifies the destination address (<https://www.w3schools.com/html/>) of the link.
- The **link text** is the visible part (Visit our HTML tutorial).
- Clicking on the link text will send you to the specified address.

URL:

- **URL** stands for Uniform Resource Locator.
- URLs (Uniform Resource Locators) or Resource Locators are unique addresses used to locate a Web page and its contents on a network serve.
- Any internet location available on server is called a web URL, web address or website.
- Each website or webpage has a unique address called URL.

Types of URL: URL gives the address of files created for webpages or other documents like an image, pdf for a doc file, etc.

There are two types of URL:

- Absolute URL
- Relative URL

What are Absolute URLs?

- Absolute URL requires you to place the entire address on the page that you link to.
- An example of an absolute URL would look like this:
- `<a href = http://www.example.com/xyz.html>`

What are Relative URLs?

- The relative URL, on the other hand, does not use the full address.
- It assumes that the page you type in is on the same site.
- An example of a relative URL would look like this:
- `<a href = "/xyz.html">`

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Local Links</h2>
```

```
<p><a href="html_images.asp">HTML Images</a> is a link to a page on this website.</p>
```

```
<p><a href="https://www.w3.org/">W3C</a> is a link to a website on the World Wide Web.</p>
```

```
</body>
```

```
</html>
```

Local Links

[HTML Images](#) is a link to a page on this website.

[W3C](https://www.w3.org/) is a link to a website on the World Wide Web.

Difference between Absolute and Relative URL:

ABSOLUTE URL	RELATIVE URL
Absolute URL requires you to place the entire address on the page that you link to.	The relative URL, does not use the full address. It assumes that the page you type in is on the same site.
It also referred to as full path.	It also referred to as non absolute path.
Absolute url consumes more time to download files.	Relative url consumes less time to download files.
Example: 	Example:

Target Attribute

- The target attribute specifies where to open the linked document:
- `<a href="https://www.w3schools.com" target="_blank">Visit W3Schools</a>`

Syntax

```
<a target="_blank|_self|_parent|_top|framename">
```

Attribute Values

Value	Description
<code>_blank</code>	Opens the linked document in a new window or tab
<code>_self</code>	Opens the linked document in the same frame as it was clicked (this is default)
<code>_parent</code>	Opens the linked document in the parent frame
<code>_top</code>	Opens the linked document in the full body of the window
<i>framename</i>	Opens the linked document in a named frame

Anchor tag: , Linking To document in folders

- Within the same folder, just use the file name:

- `<a href="thefile.html">my link</a>`

- Within the parent folder's directory:

- `<a href="../thefile.html">my link</a>`

- Within a sub-directory:

- `<a href="subdir/thefile.html">my link</a>`

Anchor Element in HTML

- The anchor element is used to create hyperlinks between a source anchor and a destination anchor.
- The source is the text, image, or button that links to another resource
- The destination is the resource that the source anchor links to.

The Anchor Element

- The anchor element tag is the letter “a” surrounded by angle brackets like this: `<a>`.
- Both the opening and closing attributes are required, and all of the content between the tags makes up the anchor source.

If we want to use just a single word as an anchor, we wrap just that one `<a>word</a>` in anchor tags.

We could also wrap an entire paragraph in anchor tags if we wanted the entire paragraph to link somewhere.

```
<!--We can also use an image as an anchor element-->  
<a></a>
```

Anchor tag to specific section within the document

- HTML bookmarks are used to allow readers to jump to specific parts of a Web page.
- Bookmarks can be useful if your webpage is very long.
- To make a bookmark, you must first create the bookmark, and then add a link to it.
- When the link is clicked, the page will scroll to the location with the bookmark.

Demo for bookmark

```
<!DOCTYPE html>
<html>
<body>

<p><a href="#C1">Jump to Chapter 1</a></p>
<p><a href="#C4">Jump to Chapter 4</a></p>

<h2>Chapter 1</h2>
<p>This chapter explains ba bla bla</p>

<h2>Chapter 2</h2>
<p>This chapter explains ba bla bla</p>

<h2>Chapter 3</h2>
<p>This chapter explains ba bla bla</p>

<h2 id="C4">Chapter 4</h2>
<p>This chapter explains ba bla bla</p>

<h2>Chapter 5</h2>
<p>This chapter explains ba bla bla</p>

<h2>Chapter 6</h2>
```

[Jump to Chapter 1](#)

[Jump to Chapter 4](#)

Chapter 1

This chapter explains ba bla bla

Chapter 2

This chapter explains ba bla bla

Chapter 3

This chapter explains ba bla bla

Chapter 4

This chapter explains ba bla bla

HTML Tag:

Definition and Usage

The tag is used to embed an image in an HTML page.

Images are not technically inserted into a web page; images are linked to web pages. The tag creates a holding space for the referenced image.

The tag has two required attributes:

- src - Specifies the path to the image
- alt - Specifies an alternate text for the image, if the image for some reason cannot be displayed

Note: Also, always specify the width and height of an image. If width and height are not specified, the page might flicker while the image loads.

```

```